Data Quality Monitoring Plan - MediLink Health Network

The purpose of the data quality monitoring plan is to ensure that data within the MediLink Enterprise Data Platform remains accurate, complete, consistent, valid, unique, and timely after the initial cleansing process. The plan establishes routine monitoring activities, performance indicators, reporting frequencies, responsible personnel, and corrective actions.

1. Monitoring activities and schedule

Monitoring activity	Frequency	Responsible person	Corrective action
Validate incoming patient data	At every ingestion	Data Custodian	Reject or quarantine records that fail validation.
Check missing values and invalid formats	Daily	Database Administrator	Generate exception reports and correct verified errors.
Monitor duplicate patient records	Daily	Data Steward	Investigate potential duplicates and approve appropriate record merges.
Review `v_dq_scorecard`	Weekly	Data Governance Lead	Investigate indicators that fall below established targets.
Review data quality by facility	Monthly	Facility Data Officers	Identify recurring errors and provide corrective training.
Review validation rules and quality trends	Quarterly	Data Governance Committee	Update controls and revise procedures where necessary.

2. Key performance indicators (KPIs)

MediLink will use the following indicators to evaluate data quality.

Indicator	Measurement method	Target
Phone-number completeness	Records with valid, non-missing phone numbers ÷ eligible patient records × 100	≥ 97%
Phone-number validity	Records with correctly formatted phone numbers ÷ records with supplied phone numbers × 100	≥ 97%
Mandatory-field completeness	Records with all required fields populated ÷ total eligible records × 100	≥ 98%
Duplicate resolution	Identified duplicate cases resolved or formally referred ÷ total identified cases × 100	100%

Referential integrity	Accepted records with valid foreign-key references	100%
Invalid categorical values	Accepted records containing unapproved categorical values	0
Timely monthly reporting	Monthly reports submitted by the fifth day ÷ reports due × 100	100%

These targets are proposed operational thresholds except for the 97% phone-number completeness requirement and the fifth-day reporting deadline, which are specified project requirements. MediLink should approve the additional thresholds through its data governance process.

3. Monitoring and escalation procedure

The monitoring process will follow these steps:

1. Collect: Run validation rules against incoming data and the central database.
2. Measure: Calculate quality indicators and update the `v_dq_scorecard` view.
3. Compare: Compare measured results against approved targets.
4. Investigate: Identify the affected facilities, records, and likely causes of deviations.
5. Correct: Assign exceptions to the responsible data custodian or data steward.
6. Verify: Rerun the validation checks to confirm that the correction resolved the issue.
7. Report: Document the results, unresolved exceptions, corrective actions, and trends for management review.

Critical problems, such as unauthorised changes to patient records or suspected exposure of sensitive data, must be escalated immediately through the security incident response process rather than waiting for the next scheduled review.

4. Reporting and accountability

The Database Administrator will maintain technical monitoring results, while Data Stewards will manage duplicate resolution and data correction. Facility Data Officers will address recurring errors originating at their facilities. The Data Governance Lead will review the weekly scorecard and coordinate corrective actions.

Monthly reports will summarise performance by facility, outstanding exceptions, trends, and progress toward quality targets. The reports must be available in time to support MediLink's requirement to submit Ministry of Health reports by the fifth day of each month.

5. Continuous improvement

MediLink will review monitoring results quarterly to determine whether its data-quality controls remain effective. Repeated errors will trigger root-cause analysis, revisions to validation rules,

staff training, or improvements to data-entry processes. All significant corrections and approved changes to quality rules will be documented to maintain accountability and traceability.

Expected outcome: This monitoring plan will enable MediLink to detect data-quality problems early, assign responsibility for their resolution, measure compliance with established targets, and sustain improvements in the reliability of healthcare data.